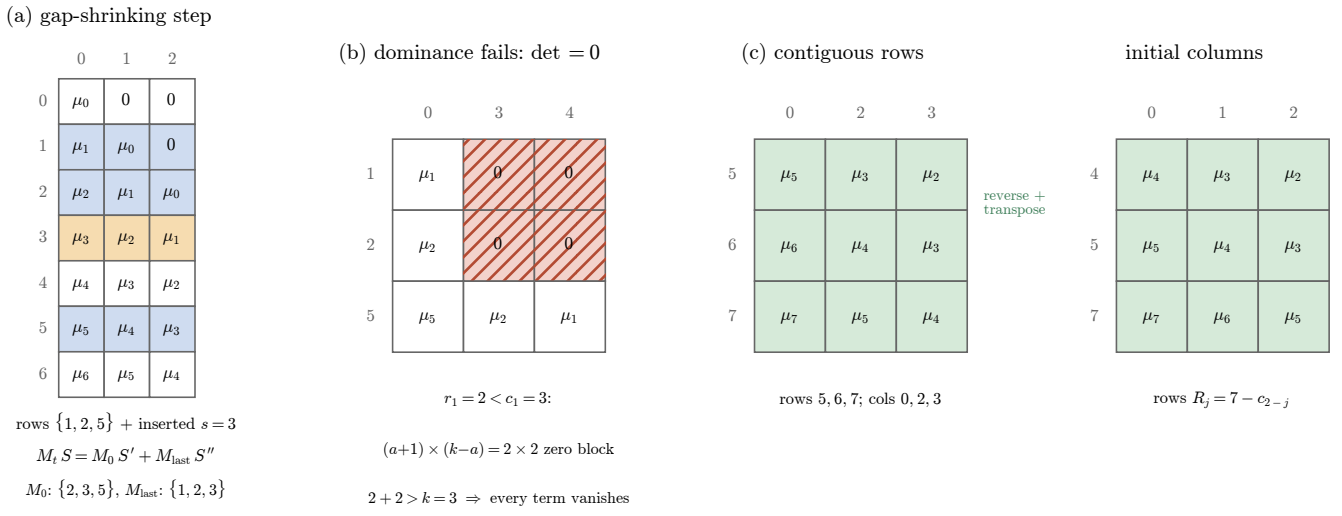
(a) gap-shrinking step
0 1 2
0 μ_0 0 0
1 μ_1 μ_0 0
2 μ_2 μ_1 μ_0
3 μ_3 μ_2 μ_1
4 μ_4 μ_3 μ_2
5 μ_5 μ_4 μ_3
6 μ_6 μ_5 μ_4
rows $\{1, 2, 5\}$ + inserted $s = 3$
$M_t\, S = M_0\, S' + M_{\text{last}}\, S''$
M_0: $\{2, 3, 5\}$, M_{last}: $\{1, 2, 3\}$
(b) dominance fails: det $= 0$
0 3 4
1 μ_1 0 0
2 μ_2 0 0
5 μ_5 μ_2 μ_1
$r_1 = 2 < c_1 = 3$:
$(a+1) \times (k-a) = 2 \times 2$ zero block
$2 + 2 > k = 3 \Rightarrow$ every term vanishes
(c) contiguous rows
0 2 3
5 μ_5 μ_3 μ_2
6 μ_6 μ_4 μ_3
7 μ_7 μ_5 μ_4
rows 5, 6, 7; cols 0, 2, 3
reverse + transpose
initial columns
0 1 2
4 μ_4 μ_3 μ_2
5 μ_5 μ_4 μ_3
7 μ_7 μ_6 μ_5
rows $R_j = 7 - c_{2-j}$